

# Crowd Sourcing Applications

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## 1 CROWD SOURCING APPLICATIONS

### 1.1 Crowd Sourcing Example

There are currently many examples of crowdsourcing applications in the world that are consistently being used everyday. However, I personally utilize Waze most often. Waze is a navigation and traffic monitoring app that helps users get from point A to point B. The app itself has a crowdsourcing aspect to it that encourages users to help other users by reporting on accidents, traffic, and even gas prices in their area while on the road.

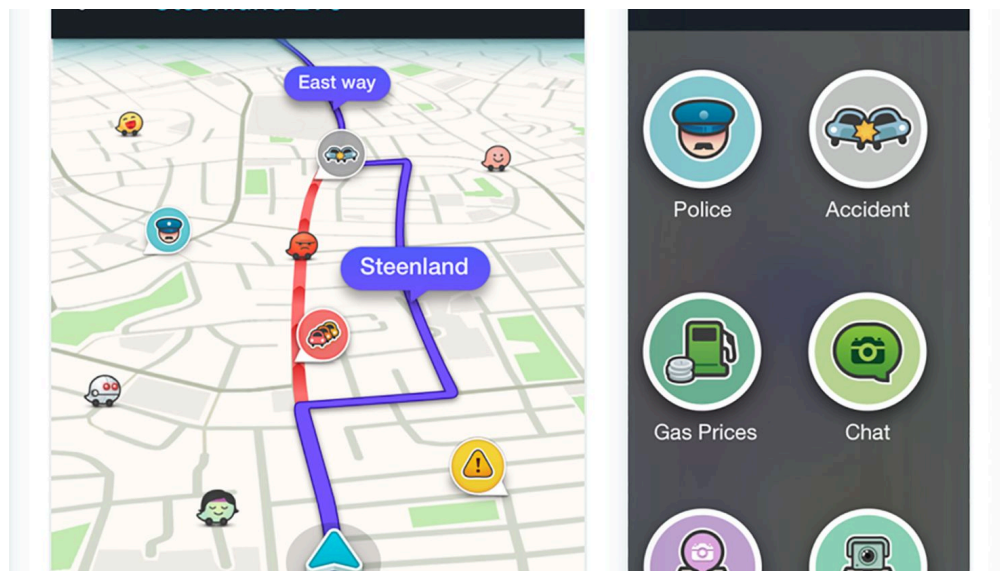


Figure 1—Image of Waze navigation and report screen. [Source](#)

As an avid user of this application, I can confidently say that Waze’s crowd sourcing in its app is a very novel way of traffic notification. Most traffic apps are able to show accidents and traffic on the road, but Waze is the only app I have found to be able to give very accurate warnings of police on the road. Even though sometimes the app is a little slow to update its accidents and traffic, the positives of the app itself far outweighs the negative, especially since the crowd sourcing is dependent on user reports. This ensures that areas of high population will most likely have accurate updates at all times. Although this feature has its pros and cons, I believe that overall this has enhanced the experience of using Waze.

With crowd sourcing becoming a popular aspect of a lot of applications, these technologies will have to be considered disruptive due to the significant changes it has already affected society. Waze is an excellent example of this due to its live updates of reports on traffic and police, due to crowd sourcing user reports. According to an article from CNN, Waze's reports on police locations have become so popular and efficient in New York City that many people will purposely use the Waze app to avoid police (Lauren, 2019). In a city as densely populated as New York City, Waze's user reports will become highly efficient considering the number of possible users. This directly alters the way police have been operating in such a prominent city. It's gotten to the point the police must ask users to stop utilizing the app. This is clearly such an impactful application that users must be persuaded to stop since it has disrupted normal society.

## **1.2 Comparison**

Crowdsourcing applications have many positive and negative aspects to it. However, it is clear that the pros far outway the cons since so many corporations have been utilizing this process for so long. The traditional counterparts of crowdsourcing do still have many situations that are more beneficial than crowdsourcing as well though. For example, the traditional counterparts have a more steady and consistent user base. No matter how popular crowdsourcing becomes, crowdsourcing is still dependent on the population of its users in a specific area. If you take Airbnb for example, if an area that a user is located in is a highly metropolitan area with little to no homeowners then the crowd sourcing will be impossible. On the other hand, hotels will become dominant in the area instead. Crowdsourcing is highly dependent on its popularity and concentration of users in an area. However, the plus side of crowdsourcing gives corporations the ability to save on time and money without having to commit to full time employees. Personally, I would suggest that new generations of crowd sourcing services have certain waves of investments in specific crowd sourcing individuals or objects that have a higher chance of return. This will ultimately turn crowd sourcing of the corporation into a seasonal transition, but this will give more opportunities for consistent growth as long as there is sufficient investments into profitable markets or individuals.

## **1.3 Human Vs Machine**

Ultimately, a machine is still a machine and humans are able to do many things better than a machine. The main thing a human is able to do better than a machine is to think outside of the box. If a machine runs into a problem that can't be solved, it will eventually crash. However, a human is able to recognize these problems and find ways around them. In addition, a machine is not able to replicate a human's senses. As long as the industry involves anything that requires taste, smell, touch, sight, and hearing, a human will always be better than a machine's input. Finally, a machine will never be able to think for itself. It does not matter how complex a

machine is, humans are the ones who innovate and create new things. Machines are able to do many things well and redo those things very well, but there are many things machines will never be able to match up to humans.

Personally, I believe that the next new crowdsourcing application will be revolving around jobs. Currently, the job market is suffering severely due to all the issues of Covid over the past few years. The repercussions are still being felt today and many people are struggling to find jobs. I believe crowdsourcing applications can be utilized in this aspect to assist people in being able to do jobs within and not within their field. The crowd sourcing is able to target a large demographic of people looking for jobs to hire so that getting a job becomes easier.

## **2 COLLABORATIVE FILTERING**

### **2.1 User-User CF Systems**

User based collaborative filtering fundamentally tries to group users together through similarities of each person so that a product liked by one person will most likely be liked by a person with similar interests. In this case, the best example would be music. There are many genres of music and it is possible to like any type of music. However, different nationalities and demographics of people tend to like similar types of music. Another case in which user based CF systems are effective are occupancy products. People in similar jobs will more likely than not use similar products. Scientists will tend to use tools like beakers and test tubes, and cooks will tend to use pots and pans. A final example of this case being effective are patients suffering from similar diseases. Just like how people can have different symptoms from the same disease, there can be many different medicines for the same disease. A user based CF system will be able to suggest medicines based on user related symptoms.

There are many cases in which user based CF systems are very effective. However, there are also situations where this is not true as well. For example, families would seemingly be grouped together based on users. However, more likely than not, just because individuals are family members does not mean that every suggestion would be good. Another example would be students. There are many types of students that study differently even though they are all students. Hypothetically, a notebook would be a good suggestion for a lot of students; however, there are many students who never use notebooks in their studies. The last example is an all encompassing example in which a user based CF system is just started. In this situation, the system would have no information on any user and would not be able to give proper suggestions until an adequate amount of data has been built up. If I were to design a user based collaborative filtering system, I would personally go around the first and second case by reducing the broadness of the system and increase the similarity conditions before offering up suggestions.

However, to handle the last situation, I would have to hold off on using the system at all until an adequate amount of data is gathered for the recommendation to be properly accurate.

## 2.2 Item-Item CF Systems

Item based collaborative systems are slightly different to user based collaborative systems in how the aforementioned failure cases would affect the system. A family with proper buying habits would not be affected by an item based CF system since the recommendations would typically be in line with what the family already buys. In addition, since an item based CF system goes off of an individual's buying habits for recommendations, the students' case is also unaffected by this situation. On the other hand, an item based CF system is also vulnerable in the same situation as a user based CF system when the system is first implemented so that the system itself does not have an adequate amount of data for a proper recommendation.

## 2.3 Comparison

User based CF systems and item based CF systems both have their advantages and disadvantages. Although they both are able to give recommendations, the process is different for each one. User based CF systems utilizes the user's patterns for a suggestion, and item based CF system utilizes similarities between items for a suggestion. Ultimately, both user based and item based systems are completely situational in which one is better than the other. User based CF systems are better when the user wants to find items that best match the user themselves, while item based CF systems are better when the user wants to find what others like best.

# 3 CONTENT BASED RECOMMENDER SYSTEMS

## 3.1 Content Based Recommendation

Crunchyroll is an online video streaming service dedicated to anime, which I use very often. Crunchyroll utilizes a content based recommendation system on its website to offer the user more anime and shows that the user would like based on what the user has watched already.

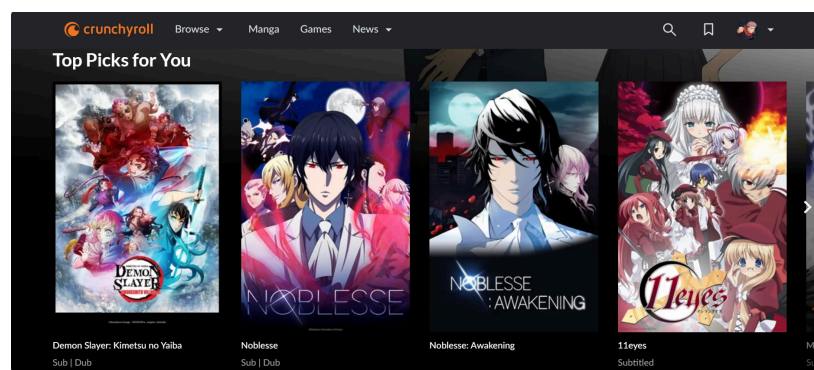


Figure 2—Image of Crunchyroll's Top Picks for You. [Source](#)

Personally, I would agree that there are aspects of this content based recommendation that I like and dislike. What I like about this system is that the recommendations are directly tailored to me personally based on my usual interests. This means that the recommendations on the list are highly likely to be interesting to me. However, what I dislike about this is that this list can be easily thrown off due to shows that I watched but did not enjoy. Since the recommendation does not take any user reports or input into account, there is a chance that the list may include suggestions that do not align with my interests.

### 3.2 CF Based Recommender System

Amazon is one of the most popular ecommerce websites in the world, and I personally have used Amazon for many years. Amazon utilizes both user based and item based CF systems on its website; however, I will only be discussing its item based CF system for this example. Amazon's wide array of products gives users multiple options whenever they are looking for a specific product. In addition, Amazon has its own personal recommendation based on other users displayed to the user whenever he or she is looking at products, known as badges.



Figure 3—Image of Amazon badges. [Source](#)

Personally, I really like the item based CF system that Amazon utilizes. Since Amazon often offers a very large selection of products based on what the user is looking for, the CF system recommendation is a quick way of having a benchmark in quality that the user can use as a reference before looking at other similar products for comparison. What I dislike about this recommendation service is that since the turnover rate of Amazon products tends to be very high, the recommendation can sometimes not accurately suggest a good product. In addition, the products that have very low review total tend to not be displayed as the recommendation compared to other products with a large number of reviews even if the reviews are bad.

### **3.3 Combination**

To combine a collaborative filtering based recommendation with a content based recommendation there would be a lot of issues that come along with it. Fundamentally, collaborative filtering and content focuses on different aspects which are hard to combine. Content based filtering focuses on the items themselves while collaborative filtering focuses on the patterns and relations between the user and items. The difficulty in combining them is how to define an appropriate split since both recommendations could essentially give different outcomes. Advantages of being able to combine both would be that the recommendation would be an especially good match for the user on all aspects of relation and the item itself, as long as both systems give the same answer. However, if each system gives different answers then besides giving both recommendations, it would be hard to differentiate which suggestion would be best to offer.

## **4 MEMORY AND MODEL BASED RECOMMENDER SYSTEMS**

### **4.1 Memory Based CF Systems**

Memory based collaborative filtering systems ultimately suffer from the curse of dimensionality because of computability. The curse of dimensionality boils down to the inability to compute a situation due to the increasing number of variables in the dataset. This is why memory based collaborative filtering systems are affected by this curse, because it is impossible to be able to calculate all of the variables that occur in real life for the system to give a perfect recommendation. There are simply too many variables surrounding a user for memory based CF systems to not be affected by the curse. Model based CF systems on the other hand are not bound by this curse due to the nature of how the recommendation is determined. The model used in the system takes the known data and trains the model for that certain situation, which is then retrained on occasion to have any changed data taken into account. This allows the model based CF systems to have educated recommendations based on the model instead of all of the variables like memory based CF systems.

### **4.2 Model Based Recommender Systems**

The specific limitations of the model based recommender systems can be seen in the uses of resources whenever the model is being trained and the accuracy of the models. Model based recommender systems utilize a model that needs to be regularly retrained on any new or updated data. This process can be time and resource consuming which other systems do not have. In addition, since the system utilizes a model that is trained on a certain amount of data, the model is not using all of the information that is available to the system. Similarly to model based recommender systems, memory based CF systems also have problems regarding accuracy and

efficiency. Since memory based CF systems take in the entirety of the dataset, if the memory system's dataset from the user has an overly abundant number of variables then this would lead to excess time used since each prediction requires reading in all the data again. Also, since all of the information from the user is taken into account, the opposite of the accuracy problem in the model based recommender system occurs, which is the overfitting of data. This leads to possible inaccuracies in the recommendations to the user.

### 4.3 New Application Domain

I believe that collaborative filtering based recommendations could really benefit the education system, specifically applying for colleges and universities. When I was applying for colleges during high school, I had no idea what major I would like, let alone what college I wanted to apply for. Collaborative filtering based recommendations could help the user get recommendations on what colleges through item based CF systems. Item based CF systems would be able to focus on the colleges themselves to give the user recommendations. At the very worst, the user would be able to take the recommendations to use as guidelines in his or her own search for colleges and universities. This CF system could also assist students in picking classes once he or she has entered college as well. I believe that collaborative filtering based recommendations can have a very strong impact on students and their education if given the chance.

## 5 REFERENCES

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